End Game Day — All Remaining Skills

Sun 8 Nov 2026 (exam eve) · Taught by Theo prof 2 · Skills **1.3.1–1.3.4**, **2.3.1–2.3.5**, **2.5.1–2.5.6**, **4.3.1–4.3.6**, **5.1.1–5.1.9**, **5.2.1, 5.2.2, 5.2.4** · After today all 98 official skills have been taught

Today in one paragraph
Check and shape data **before** it reaches a model (Glue Data Quality, Data Wrangler, Lambda checks, Textract/Transcribe/BDA, Comprehend, query rewriting, the Converse format). Bolt GenAI onto existing systems with **adapters, events and webhooks**, secure it with **federation, RBAC/ABAC and metadata filters**, keep regulated data in place with **Outposts / Wavelength / PrivateLink**, and ship through **CI/CD with GenAI tests and a central gateway**. Monitor four layers (operational, FM usage, quality, business) plus tracing and audit, and evaluate with **several metrics, several judges and several moments**.

Where to buy reliable practice exams

- **AWS Skill Builder** (skillbuilder.aws) — search 'AIP-C01': official practice question set and official practice exam. Written by AWS; start here.
- **Tutorials Dojo** (portal.tutorialsdojo.com) — AIP-C01 practice exams (~$14.99, timed + review modes, explanations) and a free 30-question sampler.
- **Udemy** — many AIP-C01 practice courses of uneven quality; only recent, highly rated ones.
- Avoid 'exam dump' sites: often wrong, and using them breaks the AWS Certification Agreement.

Part 1 — Data pipelines, input quality, input troubleshooting

```
RAW DATA (S3: PDFs, CSVs, recordings, images)
  | S3 event / EventBridge
1 VALIDATE --bad--> s3://.../quarantine/ + metric RejectedDocs + alarm
2 CONVERT  multimodal -> text/structured (Textract, Transcribe, BDA, SageMaker Processing)
3 IMPROVE  normalise, enrich, reformat (Lambda, Comprehend, Bedrock)
4 FORMAT   for the target (Converse messages, JSONL batch, endpoint payload)
  -> KB sync / fine-tuning dataset / batch inference
```

Data validation (1.3.1)

Tool	What it is	Exam cue
AWS Glue Data Quality	Rules in DQDL (Completeness, Uniqueness, ColumnLength…) on Catalog tables or in Glue jobs; results to CloudWatch / EventBridge to stop the pipeline	Tabular data at scale, rule-based, no custom code
SageMaker Data Wrangler	Visual data prep; Data Quality and Insights report (missing, duplicates, outliers, imbalance)	Data scientists preparing a training / fine-tuning set visually
Lambda checks	Per-file custom checks (parseable, language, size, PII) → quarantine prefix	Rules not enough; per-document logic
CloudWatch metrics	ValidDocs / RejectedDocs / QualityScore + alarm	Turn validation into monitoring

Example: 30% of nightly product descriptions arrive empty → Completeness rule fails → job stops before the KB sync → the chatbot never answers 'no information available'.

Multimodal processing (1.3.2)

Input	Service
Scanned docs, forms, tables	Amazon Textract (OCR + key-value pairs + tables)
Speech: calls, meetings	Amazon Transcribe (timestamps, speaker diarization, custom vocabulary, PII redaction)
Mixed docs/images/audio/video → your schema	Bedrock Data Automation (blueprints); also a KB parser
Understand images/video with an FM	Bedrock multimodal models (Claude image input, Amazon Nova)
Search images by meaning	Titan Multimodal Embeddings
Custom heavy preprocessing at scale	SageMaker Processing jobs (your container, managed batch compute)

Formatting inputs (1.3.3)

- **Converse**: one format for all models — messages [{role user|assistant, content blocks}], starts with user, alternates; system prompt in the separate system field.
- **InvokeModel**: model-specific JSON body (Anthropic anthropic_version + messages; Titan inputText…) → switching model by changing modelId alone gives ValidationException.
- **SageMaker endpoint**: whatever the container expects, e.g. {"inputs": …, "parameters": …} + ContentType.
- **Batch inference**: JSONL, one {"recordId", "modelInput"} per line.

Improving input quality (1.3.4)

- **Lambda normalisation**: strip HTML, fix encoding, unify dates/units, remove boilerplate, de-duplicate.
- **Comprehend**: entities → chunk metadata; dominant language → routing / Translate; PII → mask before storage.
- **Bedrock reformatting**: small model fixes spelling, rewrites follow-ups into **standalone questions**, decomposes multi-part questions, converts free text to structure.

Troubleshooting content handling (5.2.1)

Symptom	Cause	Fix
ValidationException: input too long	Context window exceeded	Trim/summarise history, fewer/smaller chunks, compress, larger-context model
Stops mid-sentence, stopReason max_tokens	Output limit	Raise maxTokens, shorter format, continue call
Long-doc summary ignores the end	Truncation / lost middle	Map-reduce summarisation
Chunks very uneven	One chunk size for everything	Dynamic chunking per document type

Troubleshooting FM integration (5.2.2)

Structured error logs (request ID, model ID, Region, error, tokens); **API Gateway request validation** rejects bad requests before paying for Lambda/Bedrock; always read **stopReason** (end_turn, max_tokens, tool_use, guardrail_intervened, content_filtered).

Error	Action
ValidationException	Fix the request (shape, length) — don't retry
AccessDeniedException	Fix IAM / enable model access / KMS key policy

Error	Action
ResourceNotFoundException	Wrong model ID or Region (or use an inference profile)
ThrottlingException	Backoff + jitter; quota increase; cross-Region inference
ModelTimeout / ServiceUnavailable / ModelNotReady	Retry

Troubleshooting retrieval (5.2.4)

First call **Retrieve** alone: right chunks but wrong answer → prompt/grounding problem; wrong chunks → retrieval problem.

Symptom	Fix
All scores low after a model change	Same embedding model for docs and queries; re-embed everything
Exact codes / SKUs missed	Hybrid search
Right doc, chunk lacks context; tables split	Larger chunks, overlap, hierarchical chunking, parse tables whole
Answer drowned in noise	Smaller / semantic chunks
Relevant chunk ranked low	Reranker (or higher top-k then rerank)
New documents never appear	Ingestion job statistics / failure reasons; sync after uploads
Quality declines over months	Drift: monitor retrieval scores and 'no answer' rate; refresh content
Slow at scale	HNSW tuning, more OCUs/capacity, metadata filters

Part 2 — Enterprise integration, CI/CD, gateway, interfaces, dev tools

Enterprise connectivity (2.3.1)

```
GenAI app/agent -> API Gateway (REST, auth, throttling)
                -> Lambda ADAPTER (JSON <-> SOAP/XML)
                -> VPN / Direct Connect -> legacy system on-premises   (+ SQS buffer for fragile systems)
```

- **Event-driven loose coupling**: EventBridge rules route business events (TicketCreated) to GenAI Lambdas.
- **Data sync**: SaaS → S3 with **AppFlow**; databases continuously with **DMS CDC**; sources → KB with data source connectors and **StartIngestionJob** triggered by change events.

Adding GenAI to existing apps (2.3.2)

```
Sync:     App -> API Gateway (/summarise) -> Lambda -> Bedrock -> response
Webhook:  Sender -> API GW/Lambda (verify signature, 200 OK fast) -> SQS -> worker Lambda -> Bedrock
Events:   CRM 'CaseClosed' -> EventBridge rule -> Lambda -> Bedrock summary -> CRM
```

Secure access (2.3.3)

- Federation: **IAM Identity Center** (workforce SSO), **Cognito** user pools (sign-in, tokens) + identity pools (temporary AWS credentials), **API Gateway authorizers** (Cognito / Lambda).
- RBAC for models (InvokeModel only on approved ARNs) and data (**group claim → KB metadata filter**); ABAC with tags.
- Least privilege: resource-level ARNs, condition keys, SCP deny for unapproved models, VPC endpoints.

Cross-environment compliance (2.3.4)

Service	Use when
AWS Outposts (AWS racks in your data centre)	Data must stay on premises; low latency to on-prem systems; self-hosted open models (Bedrock is a Region service)
AWS Wavelength (AWS compute in 5G telecom networks)	Ultra-low latency to mobile/edge devices
AWS Local Zones	Low latency to a metro area without own hardware
Direct Connect / VPN + PrivateLink + Transit Gateway	Private routing between on-prem and Bedrock
Geographic cross-Region inference profile + SCP Region deny	Residency within a geography in the cloud

CI/CD and the GenAI gateway (2.3.5)

```
push -> CodePipeline
  CodeBuild: build + unit tests | security scans | GenAI tests (prompt regression, guardrail attack prompts
             schema checks, cost/latency budget)
  manual approval (prod) -> deploy (CloudFormation/CDK) -> CodeDeploy canary on Lambda alias
  -> CloudWatch alarm -> AUTOMATIC ROLLBACK      (prompts/flows: move version pointer/alias back)

Teams -> GenAI GATEWAY (API Gateway + Lambda/ECS): auth, per-team usage plans, approved-model routing/fallb
         central guardrails + PII masking, logging, token metering, cost per team -> Bedrock / other provid
```

Interfaces and developer tools (2.5.1–2.5.6)

- **2.5.1** Streaming: ConverseStream → WebSocket API or SSE / response streaming. API Gateway integration timeout 29 s by default → stream or go async (job ID + SQS). Token budgets; SDK standard/adaptive retries, backoff + jitter, fallback model.
- **2.5.2 Amplify** (declarative UI components incl. AI chat, hosting); **OpenAPI-first** contracts (API Gateway import, SDKs, agent tool schemas); **Bedrock Flows** no-code builder.
- **2.5.3** Lambda + CRM (summarise history, draft reply, write back); Step Functions document processing (BDA/Textract → classify → Choice on confidence → human task token → store); **BDA blueprints**.
- **2.5.4 Amazon Q Developer**: generate/refactor code, API suggestions, unit tests, security scans, performance tips, agentic multi-file changes. (Q Business = employee assistant over company data.)
- **2.5.5** Strands, Agent Squad, Step Functions agent patterns, Bedrock prompt chaining (recap).
- **2.5.6** Logs Insights over prompts/responses; X-Ray across FM calls; Q Developer for GenAI error patterns.

Part 3 — Monitoring and observability (Task 4.3)

```
4 BUSINESS     deflection, CSAT, conversions, cost per resolved case   (is it worth it?)
3 QUALITY      hallucination rate, relevance, ratings, drift           (is it RIGHT?)  GenAI-specific
2 FM USAGE     tokens, cost, guardrail blocks, stopReason mix          (how is it used?) GenAI-specific
1 OPERATIONAL  latency, errors, throttles                              (is it UP?)
  + TRACING (X-Ray / OpenTelemetry, AgentCore Observability) + AUDIT (CloudTrail, invocation logs)
```

Skill	Key points
4.3.1 Holistic	AWS/Bedrock metrics (Invocations, InvocationLatency, Client/ServerErrors, Throttles, Input/OutputTokenCount); X-Ray performance tracing; invocation logs + AgentCore traces for FM interactions; business custom metrics on CloudWatch dashboards / QuickSight.
4.3.2 KPIs	Token usage, prompt effectiveness (pass rate per version), hallucination rate (grounding / judge on samples), quality, response drift. **CloudWatch anomaly detection** for time-varying bursts; **Cost Anomaly Detection** for unusual spend; invocation logs for detail; benchmarks per version.

Skill	Key points
4.3.3 Insights	Dashboards; **AWS Config** compliance rules + Security Hub; forensics with CloudTrail / **CloudTrail Lake** SQL, invocation logs in **S3 Object Lock**, correlation IDs; user interaction and behaviour-pattern tracking (refusal, guardrail-intervention rate, length, topics).
4.3.4 Tools	Call patterns, per-tool latency/errors, tool-calling and multi-agent hand-off traces, usage baselines → anomaly alarms (loops).
4.3.5 Vector store	Latency, request rate, indexing latency, OCUs; scheduled index optimisation (HNSW ef_search, segment merge, pgvector VACUUM/ANALYZE/reindex, full re-index after embedding change); data quality (ingestion statistics, duplicates, stale and orphaned vectors).
4.3.6 Failure modes	Golden datasets → hallucinations/regressions; output diffing (semantic similarity) → inconsistency/drift; reasoning path tracing → logic errors; pipeline: logs → S3 → Athena / Firehose → OpenSearch, Lambda judge on samples → CloudWatch metrics.

Part 4 — Evaluation (Task 5.1)

```
WHAT: relevance, correctness, faithfulness, completeness, consistency, fluency, harm, retrieval, agent
WHO:  code (exact checks, BLEU/ROUGE/BERTScore) | LLM-as-a-judge | humans (users, annotators)
WHEN: design (choose) | CI/CD (quality gate) | rollout (A/B, canary, shadow) | production (continuous)
```

Skill	Key points
5.1.1 Metrics	Relevance, factual accuracy, faithfulness, completeness, consistency, fluency, harmfulness. BLEU/ROUGE = word overlap; BERTScore = meaning.
5.1.2 Systematic	**Bedrock Evaluations**: automatic, LLM-as-a-judge, human; bring your own responses; compare jobs. A/B = compare (business outcomes); canary = safe rollout; multi-model; token efficiency, latency-to-quality, cost per success.
5.1.3 User-centred	Thumbs, stars, comments linked to response IDs; implicit signals (rephrase, regenerate, escalate); annotation with SageMaker Ground Truth / A2I / Bedrock human eval → new golden cases.
5.1.4 QA process	Scheduled continuous evaluation in production; regression tests on every change; automated quality gates that fail deployments.
5.1.5 Multi-perspective	RAG evaluation + LLM-as-a-judge (calibrated, different judge model) + human sample.
5.1.6 Retrieval	Context relevance, context coverage; hit rate / recall@k, precision@k, MRR; Retrieve latency p50/p95. KB evaluation: retrieve-only or retrieve-and-generate (faithfulness, citation precision/coverage).
5.1.7 Agents	Task completion rate, tool selection/argument correctness, steps/tokens/cost per task, reasoning quality from traces; managed agent evaluation on scenario sets incl. edge cases.
5.1.8 Reporting	QuickSight (business), CloudWatch dashboards (ops), Athena over results; scheduled Lambda reports via SNS/SES; model comparison charts; match the audience.
5.1.9 Deployment validation	CloudWatch Synthetics canaries running scripted conversations; hallucination and **semantic drift** vs baseline; consistency checks; **shadow testing** (no user impact); canary with automatic rollback.

End Game quiz — 20 / 20 (100%)

Q	Level	Skill	Answer	Why (and the trap)
1	Hard	1.3.1	C ✓	Glue Data Quality (DQDL) + EventBridge halt. Lambda = custom code; Data Wrangler = manual.
2	Medium	1.3.2	B ✓	One managed service, many file types, defined schema → BDA blueprints.
3	Hard	1.3.3	B, D ✓	Converse = one format; roles must alternate starting with user. Retrying a ValidationException never helps.
4	Hard	1.3.4	D ✓	Follow-ups lack context → rewrite to standalone question; index untouched.
5	Medium	5.2.1	C ✓	max_tokens with small input = output limit, not context window.
6	Hard	5.2.4	A, D ✓	Codes → hybrid search; query/doc embedding mismatch → re-embed with one model.
7	Medium	2.3.1	D ✓	Throttled API Gateway + adapter over private link; nightly copy isn't current.
8	Hard	2.3.2	B, E ✓	Acknowledge within 5 s, do the 10–30 s work asynchronously.
9	Hard	2.3.3	C ✓	Federation + department claim → metadata filter; prompts and guardrails don't enforce access.
10	Medium	2.3.4	A ✓	Data must stay in the factory → Outposts with a self-hosted model.
11	Hard	2.3.5	B, D, F ✓	Gateway + per-team cost attribution + CI/CD with canary rollback.
12	Medium	2.5.1	D ✓	Streaming fixes both the 29 s timeout and the perceived wait.
13	Medium	2.5.2/2.5.4	B, D ✓	Amplify UI for the front-end; Q Developer for code work (not Q Business).
14	Hard	2.5.3	A ✓	Two-day human wait → Standard workflow + task token; Express max 5 min.
15	Hard	4.3.2	C, E ✓	Time-varying metric → anomaly detection; unusual spend → Cost Anomaly Detection.
16	Medium	4.3.3	C ✓	Exact response + tamper-proof for 7 years → invocation logs in S3 Object Lock.
17	Hard	4.3.4/4.3.6	C, E ✓	Cost up, answers fine → tool-call baselines (loops); wrong conclusion → reasoning traces.
18	Hard	5.1.2/5.1.9	A, E ✓	No user impact → shadow; gradual + rollback → canary with alarms.
19	Medium	5.1.6	A ✓	KB evaluation retrieve-only: context relevance + coverage.
20	Hard	5.1.3–5.1.5	B, D, E, F ✓	Judge for scale, experts for tone, users for learning, gate for deployments.

Glossary — new terms

Term	Meaning
AWS Glue Data Quality / DQDL	Rule-based data checks in Glue / the rule language (Completeness, Uniqueness…).
SageMaker Data Wrangler	Visual data preparation with a data quality and insights report.
Quarantine prefix	S3 location where invalid files are moved for review.
Multimodal	Data beyond plain text: images, scans, audio, video.
Amazon Textract	OCR service extracting text, key-value pairs and tables from documents.
OCR (Optical Character Recognition)	Reading text from images.
Amazon Transcribe	Speech-to-text with diarization, custom vocabulary and PII redaction.
Speaker diarization	Labelling who spoke when in a recording.
Bedrock Data Automation (BDA) / blueprint	Managed extraction from documents and media / your field schema for it.
SageMaker Processing job	Managed batch compute running your own preprocessing container.
Titan Multimodal Embeddings	Embedding model placing images and text in one vector space.
JSONL	File with one JSON object per line (batch inference input).
Query rewriting / decomposition	Turning a follow-up into a standalone question / splitting a complex question.
Map-reduce summarisation	Summarise chunks separately, then summarise the summaries.
Dynamic chunking	Choosing chunking strategy and size per document type.
API Gateway request validation	Schema checks that reject malformed requests before the backend runs.
Adapter (Lambda)	Translator between a modern JSON API and a legacy protocol such as SOAP.
Direct Connect / Site-to-Site VPN	Dedicated private line / encrypted tunnel between on-premises and AWS.
Loose coupling	Systems interact through events, so each can change or fail independently.
Amazon EventBridge	Event bus routing events to targets by rules.
Amazon AppFlow	Managed no-code data transfer from SaaS apps to AWS.
AWS DMS / CDC	Database Migration Service / Change Data Capture streaming only changed rows.
Webhook	HTTP callback a third-party system sends when an event happens.
IAM Identity Center	Workforce single sign-on to AWS accounts and roles.
Amazon Cognito user / identity pool	App user sign-in and tokens / exchange for temporary AWS credentials.
Lambda authorizer	Custom token check protecting an API Gateway API.
RBAC / ABAC	Access by role / access by attributes such as tags.
AWS Outposts	AWS-managed racks running AWS services in your own data centre.
AWS Wavelength	AWS compute inside telecom 5G networks for ultra-low latency.
AWS Local Zones	Small AWS sites near large cities.
Transit Gateway	Hub connecting many VPCs and on-premises networks.
CI/CD	Continuous Integration / Continuous Delivery: automated build, test, deploy.
CodePipeline / CodeBuild / CodeDeploy	Pipeline orchestration / build and test runs / gradual deploys with rollback.

Term	Meaning
GenAI gateway	Central layer all apps use to reach models: auth, quotas, routing, guardrails, logging, cost.
Usage plan	API Gateway per-client throttling and quota tied to API keys.
SSE (Server-Sent Events)	One-way server-to-browser stream over HTTP.
AWS Amplify / Amplify UI	Front-end framework and hosting / declarative ready-made UI components.
OpenAPI specification	Machine-readable API contract used for API-first development.
Amazon Q Developer	AI coding assistant: generate, refactor, test, explain, optimise code.
Amazon Q Business	Enterprise assistant answering employees from company data.
KPI	Key Performance Indicator.
CloudWatch anomaly detection	Learned expected band for a metric; alarm when outside it.
AWS Cost Anomaly Detection	Billing feature alerting on unusual spend patterns.
AWS Config	Continuous checks of resource configuration against rules.
CloudTrail Lake	Queryable (SQL) store of CloudTrail audit events.
S3 Object Lock (WORM)	Write once, read many: objects can't be changed or deleted for a period.
Correlation ID	Request ID carried through every log line to trace one request end to end.
Output diffing	Comparing outputs for the same input across runs or versions.
Reasoning path tracing	Inspecting each reasoning/tool step to find where logic fails.
BLEU / ROUGE / BERTScore	Word-overlap metrics / meaning-based similarity metric.
Faithfulness	Every claim in the answer is supported by the provided context.
A/B test / canary / shadow test	Compare configs on split users / small % rollout / mirror traffic without showing results.
SageMaker Ground Truth / Amazon A2I	Labelling workforce service / human review loops.
Quality gate	Pipeline step that fails a deployment below quality thresholds.
Context relevance / context coverage	Retrieved chunks are relevant / contain the needed information.
Hit rate (recall@k) / precision@k / MRR	Right chunk in top k / share of top k relevant / rewards rank 1.
Task completion rate	Share of agent tasks fully and correctly finished.
CloudWatch Synthetics canary	Scheduled script acting as a synthetic user against an endpoint.
Semantic drift	Answers' meaning shifting versus a baseline for the same questions.

Next
Read the Final Exam Cheat Sheet (decision rules, trick cues, mistakes, consolidated glossary), then practice questions until 12:30 PM. Exam 1:30 PM IST.